Question	Answer	Marks
2(a)	force × <u>perpendicular</u> distance (of line of action of force to / from the point)	**B1**
2(b)(i)	$150 \times 9.81 \times 4.5$ **or** $90 \times 9.81 \times 4.5$ **or** $3.0 \times 9.81 \times m$	**C1**
	$(150 \times 9.81 \times 4.5) = (90 \times 9.81 \times 4.5) + (3.0 \times 9.81 \times m)$	**C1**
	$m = 90$ kg	**A1**
2(b)(ii)	Young modulus = σ / ε **or** $F / A\varepsilon$ **or** FL / Ax	**C1**
	Area of wire $= \pi \times (1.8 \times 10^{-3} / 2)^2$ $= 2.5 \times 10^{-6}$	**C1**
	So Young modulus $= ((90 \times 9.81) / \pi \times (9.0 \times 10^{-4})^2\ ) / 1.2 \times 10^{-3}$ $= 2.9 \times 10^{11}$ Pa	**A1**
2(b)(iii)	Moment provided by B will decrease / moment due to wire will increase	**B1**
	So force acting on wire will increase (Young's modulus and area remain constant) and the strain will increase	**B1**

Question	Answer	Marks